Skip to content

- Post category:Class 7 Notes

Class 7 Math Notes In English Chapter – 08 Comparing Quantities available here in English as well as in English – you can Download chapter wise notes, Class 7 Math Notes by Notes Street is prepared by notesstreet Team to help you understand the every part of topic in depth. The notes of Class 7 Hindi And English PDF is your best choice of study material If you are looking out to score high marks. Notes of Class 7 Math Notes In English Chapter – 08 Comparing Quantities are prepared in accordance with NCERT Syllabus and CBSE guidelines from expert teachers. Notes are as per the NCERT syllabus and guidelines. Each topic in our Chemistry notes are explained as per the exam structure. Along with easy and best explanation we have provided coloured diagrams, images so that topic can be understood in first attempt.. Notes are very useful to score high and colorful diagram is given so it's eye catching and easy to understand. if you want to premium notes Then **Click Here**

Class 7 Math Notes In English Chapter – 08 Comparing Quantities

---

Ratio and Proportion

1. Ratio is a method of comparing two quantities of same kind by division. Symbol used to write ratio is ':' and is read as SS 'is to'. Ratio is always expressed in its simplest form.

2. Two or more ratios are equivalent if their corresponding fractions are equivalent.

3. When two ratios are equal, they are said to be in proportion. Symbol for proportion is :: and is read as 'as to'.

4. If two ratios are equal or to be in proportion, their product of means should be equal to the product of extremes.

5. That is, if a : b : : c : d then ad = bc.

6. The method to find out the value of one unit/item which in turn is used to find the value of required number of units/items is called unitary method.

Percentages

1. Percentages are numerators of fractions with denominator 100. The symbol '%' is used for percent and it indicates multiplication with 1/100.

For example: 16% = 16/100

2. To convert a fraction into a percentage, we multiply the fraction by 100.

3. To convert a decimal into a percentage, write the decimal as fraction and then multiply the fraction by 100.

4. To convert a percentage into a fraction (decimal), drop the percent sign and divide by 100 and then reduce it into simplest form of fraction (decimal).

1. Percentage increase/ decrease = (Amount of change/Original amount) × 100%

Profit and Loss

1. The price at which an article is purchased is called the cost price (CP).

2. The price at which an article is sold is called the selling price (SP).

3. If SP > CP, then a profit is made.

Gain = (SP) – (CP)

4. If SP < CP, then a loss is incurred.

Loss = (CP) – (SP)

5.

$$\text{Gain\%} = \left\{\frac{\text{gain}\times 100}{\text{CP}}\right\}$$

$$\text{Loss\%} = \left\{\frac{\text{loss} \times 100}{\text{CP}}\right\}$$

Simple Interest

1. Whenever we borrow money from some lending sources such as banks or financial institutions etc., we have to pay some extra money for the service of lending. This extra money

depends on the sum we borrow and the period of time for which we borrow. This extra money is called the interest.

2. The money borrowed is called the principal or sum.

3. Amount = (principal + interest).

4. Interest on 100 for 1 year is called the rate per cent per annum.

5. If the interest is calculated uniformly on the original principal, it is called the simple interest.

6. If stand for principal, rate and time respectively, and SI stands for simple interest, then: SI = (P × R × T) / 100

Prices related to an item

- ➡ Prices related to an item are: (i) Selling price , (ii) Cost price
- ➡ Selling price (SP) is the price at which a product is sold out.
- ➡ Cost price (CP) is the buying price of an item.
- ➡ Profit = Selling price – Cost price
- ➡ • Loss = Cost price – Selling price
- ➡ If SP > CP, then it is profit.
- ➡ If SP = CP, then it is neither profit nor loss.
- ➡ If CP > SP, then it is loss.

Finding the profit or loss percentage

➡ Profit Percentage = $\frac{Profit}{Cost\ price} \times 100$

➡ Loss Percentage = $\frac{Loss}{Cost\ price} \times 100$

Percentages

Percentages are ratios expressed as a fraction of 100.

Percentages are represented by the symbol '%'.

Example: 20/100 = 20% and 50/100 = 50%

Comparing percentages when denominator is not 100

When a ratio is not expressed in fraction of 100, then convert the fraction to an equivalent fraction with denominator 100.

Example: Consider a fraction 3/5 Multiply the numerator and denominator by 20.

$\Rightarrow (3 \times 20)/(5 \times 20) = 60/100 = 60\%$

Converting fractions/decimals to percentages

Converting Decimals to Percentages

Given decimal: 0.44

$0.44 = 44/100 = 44/100 \times 100\% = 44\%$

Converting Fraction to percentages

Given fraction

$3/5 \times 100\% = 3 \times 20\% = 60$

Converting percentages to fractions/decimals

$0.25 = 25/100 = ¼$

$0.225 = 225/1000 = 9/40$

Estimation using percentages

Estimation can be done using percentages.

Example: What percentage of the given circle is shaded?

Solution: The given triangle consists of 8 regions, out of which 6 regions are shaded. So, the percentage of shaded regions will be

$6/8 \times 100 = \frac{3}{4} \times 100 = 75\%$

Interpreting percentage into usable data

➤ Percentages can be interpreted into useful data.

➤ Examples:

(i) 40% of Raghav's clothes are not washed.

⇒ Raghav's 40 clothes out of 100 clothes are not washed.

(ii) 30 % of students in class are infected by fever.

⇒ Out of 100 students in a class, 30 students are infected by fever.

Converting percentage to the form "how many"

Example: 200 chocolates were distributed among two children: Joe and Tom. Joe got 60% and Tom got 40% of the chocolates. How many chocolates will each get?

Solution: Total number of chocolates = 200

Joe got 60% of the chocolates = $60/100 \times 200 = 120$ Tom got 40% of the chocolates = $40/100 \times 200 = 80$

∴ Joe and Tom will get 120 and 80 chocolates, respectively.

Converting Ratios to percentages

Ratios can be expressed as percentages to understand certain situations much better.

Example: 200 chocolates were distributed among two children: James and Jacob. James got 3/5 and Jacob got 2/5 of the chocolates. What is the percentage of chocolate that each got? Solution: Total number of chocolates = 200

James got 3/5 of the chocolates = 3/5 × 100 = 60% of the total chocolates.

Jacob got 2/5 of the chocolates = 2/5 × 100 = 40% of the total chocolates

Introduction to Fractions and Ratios

Comparing Quantities: Introduction

To compare two quantities, the units must be the same.

Examples:

(i) Joe's height is 150 cm and Tom's is 100 cm.

Ratio of Joe's height to Tom's height would be Joe's height: Tom's height.

= 150 : 100 = 3 : 2

(ii) Ratio of 3 km to 30 m is 3 km : 30 m.

= 3000 m : 30 m

= 300 : 1

Ratios

Ratio is a relation between two quantities showing the number of times one value contains or is contained within the other.

Example: If there are four girls and seven boys in a class, then the ratio of number of girls to number of boys is 4 : 7.

Equivalent Ratios

By multiplying numerator and denominator of a rational number by a non-zero integer, we obtain another rational number equivalent to the given rational number. These are called equivalent fractions.

Example: $1/3 = 1/3 \times 2/2 = 2/6$ and $1/3$ are equivalent fractions.

Proportions

If two ratios are equal, then they are said to be in proportion.

Symbol "::" or "=" is used to equate the two ratios.

Example: (i) Ratios 2:3 and 6:9 are equal. They can be represented as 2 : 3 :: 6:9 or 2 : 3 = 6 : 9.

(ii) Ratios 1 : 2 and x : 4 are in proportion.

$\Rightarrow 12 = x4$

$\Rightarrow 1 \times 4 = x \times 2$

$\Rightarrow 2x = 4$

$\Rightarrow x = 2$

Finding the percentage increase or decrease

Example: Price of a book was changed from ₹20 to ₹25 in a week. Calculate the percentage increased.

Solution: Change in price = ₹25 – ₹20 = ₹5

Percentage Increased = (Change in price)/(Original price)

$= 5/20 \times 100 = 25\%$

Simple and Compound Interest

Sum / principal

The money which has been borrowed is called sum or principal.

This money can be used by the borrower for a particular time period before returning to the lender.

Example: Loan that you take from a bank is the principal.

Interest

Interest is the extra payment that a borrower should pay to the lender along with the principal.

Amount

A borrower should return the principal amount (he/she has borrowed) and the interest to the lender. This money is called amount.

⇒ Amount = Principal + Interest.

Simple Interest

Simple Interest (S.I) is the method of calculating the interest amount for some principal amount of money. Have you ever borrowed money from your siblings when your pocket money is exhausted? Or lent him maybe? What happens when you borrow money? You use that money for the purpose you had borrowed it in the first place. After that, you return the money whenever you get the next month's pocket money from your parents. This is how borrowing and lending work at home.

But in the real world, money is not free to borrow. You often have to borrow money from banks in the form of a loan. During payback, apart from the loan amount, you pay some more money that depends on the loan amount as well as the time for which you borrow. This is called simple interest. This term finds extensive usage in banking.

Simple interest (SI) is the interest charged on a borrowed money where the principal amount will be fixed for a particular time period.

Simple Interest Formula

The formula for simple interest helps you find the interest amount if the principal amount, rate of interest and time periods are given.

Simple interest formula is given as:

$$SI = PTR/100$$

Where SI = simple interest
P = principal
R = interest rate (in percentage)
T = time duration (in years)

In order to calculate the total amount, the following formula is used:

Amount (A) = Principal (P) + Interest (I)

Where,

Amount (A) is the total money paid back at the end of the time period for which it was borrowed.

The total amount formula in case of simple interest can also be written as: $A = P (1 + RT)$

Here,
A = Total amount after the given time period
P = Principal amount or the initial loan amount
R = Rate of interest (per annum)
T = Time (in years)

Simple Interest Formula For Months

The formula to calculate the simple interest on a yearly basis has been given above. Now, let us see the formula to calculate the interest for months. Suppose P be the principal amount, R be the rate of interest per annum and n be the time (in months), then the formula can be written as:

Simple Interest for n months = $(P \times n \times R)/ (12 \times 100)$

The list of formulas of simple interest for when the time period is given in years, months and days are tabulated below:

Time	Simple interest Formula	Explanation
Years	PTR/100	T = Number of years
Months	(P × n × R)/(12×100)	n = Number of months
Days	(P x d x R)/(365 ×100)	d = Number of days (non-leap year)

Simple Interest (P × n × R/100) P = Principal Amount, R = Interest rate
N = Number of years =

Example: Calculate the simple interest for 3 years when the principal amount is 200 and interest rate is 10% for 1 year.

Solution: Given: P = 200; R = 10%; T = 3 yrs

Difference Between Simple Interest and Compound Interest

There is another type of interest called compound interest. The major difference between simple and compound interest is that simple interest is based on the principal amount of a deposit or a loan whereas compound interest is based on the principal amount and interest that accumulates in every period of time. Let's see one simple example to understand the concept of simple interest.

Compound interest

Compound interest is the addition of interest to the principal sum of a loan or deposit, or in other words, interest on interest. It is the result of reinvesting interest, rather than paying it out, so that interest in the next period is then earned on the principal sum plus previously accumulated interest.

Formula

A = final amount

P = initial principal balance
r = interest rate
n = number of times interest applied per time period
t = number of time periods elapsed

Please Share This Share this content

- Opens in a new window

- Opens in a new window

- Opens in a new window

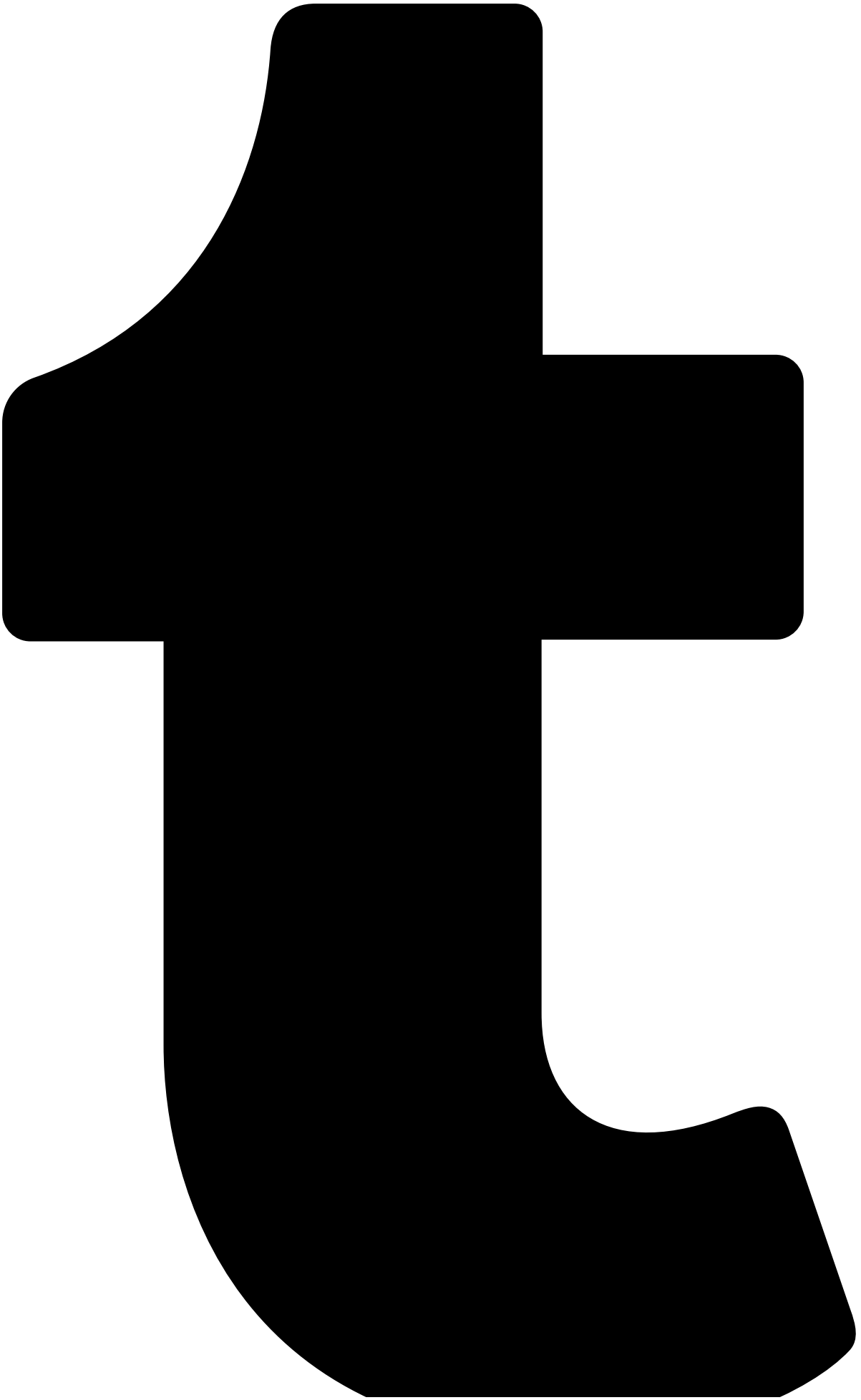
- Opens in a new window

- Opens in a new window

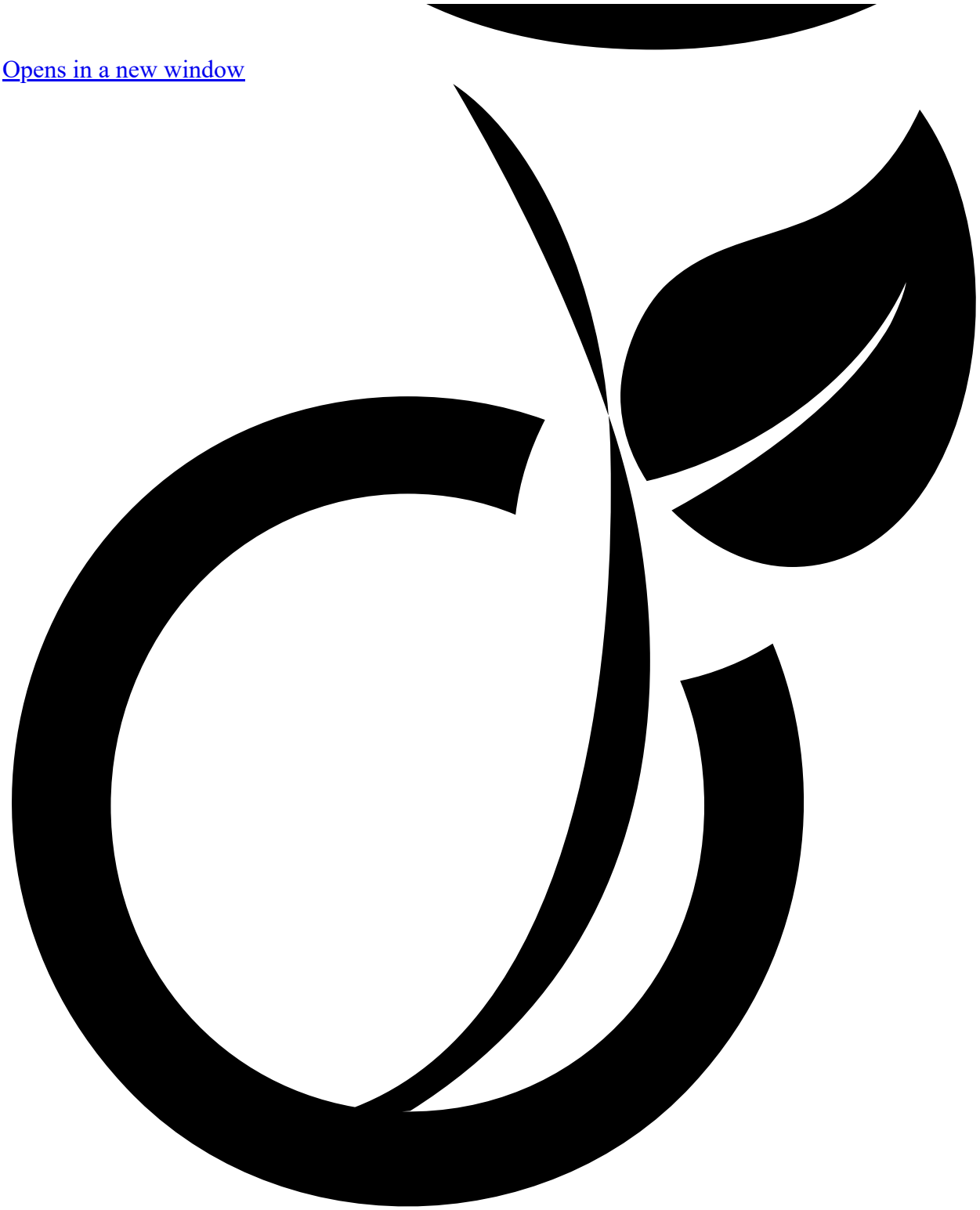
- Opens in a new window

- Opens in a new window

- Opens in a new window



- Opens in a new window



- Opens in a new window

You Might Also Like

Class 7 History Notes Ch – 04 मुग़ल साम्राज्य

March 27, 2024

Class 7 History Notes Ch – 01 Tracing Changes Through a Thousand Years

March 27, 2024

Class 7 Science

June 22, 2022